

UserStory A3 – Power Supply

(Replaced by the competency of soldering wires)



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Task according to work order

In this User Story we have to:

- Cut the power supply cable (USB side +/- 20 cm).
- Strip both ends.
- Twist the ground braid and cover it with heat-shrink tubing (leave 5mm of braid at the end to make the electrical connection).
- Strip 5mm of the +5V stranded wire.
- Cover both strands with heat-shrink tubing.

Materials needed

- Power Supply Cable (+/- 20cm)
- Small brown shrinking tube
- Large Transparent shrinking tube



Figure 1: Materials

Processing with the User story

Cutting the cable

For the first step, I had to carefully cut the cable in half so that both cables would be approximatively 10cm each.

What I paid attention to, is how I cut it but most importantly with what I cut it.

I cut it with a side-cutter, by paying attention to cut it as much as possible in half so that I could have a greater error margin.

I also already put the transparent shrinking tube on one of the cables so that after the soldering is done, I would optimize my work.

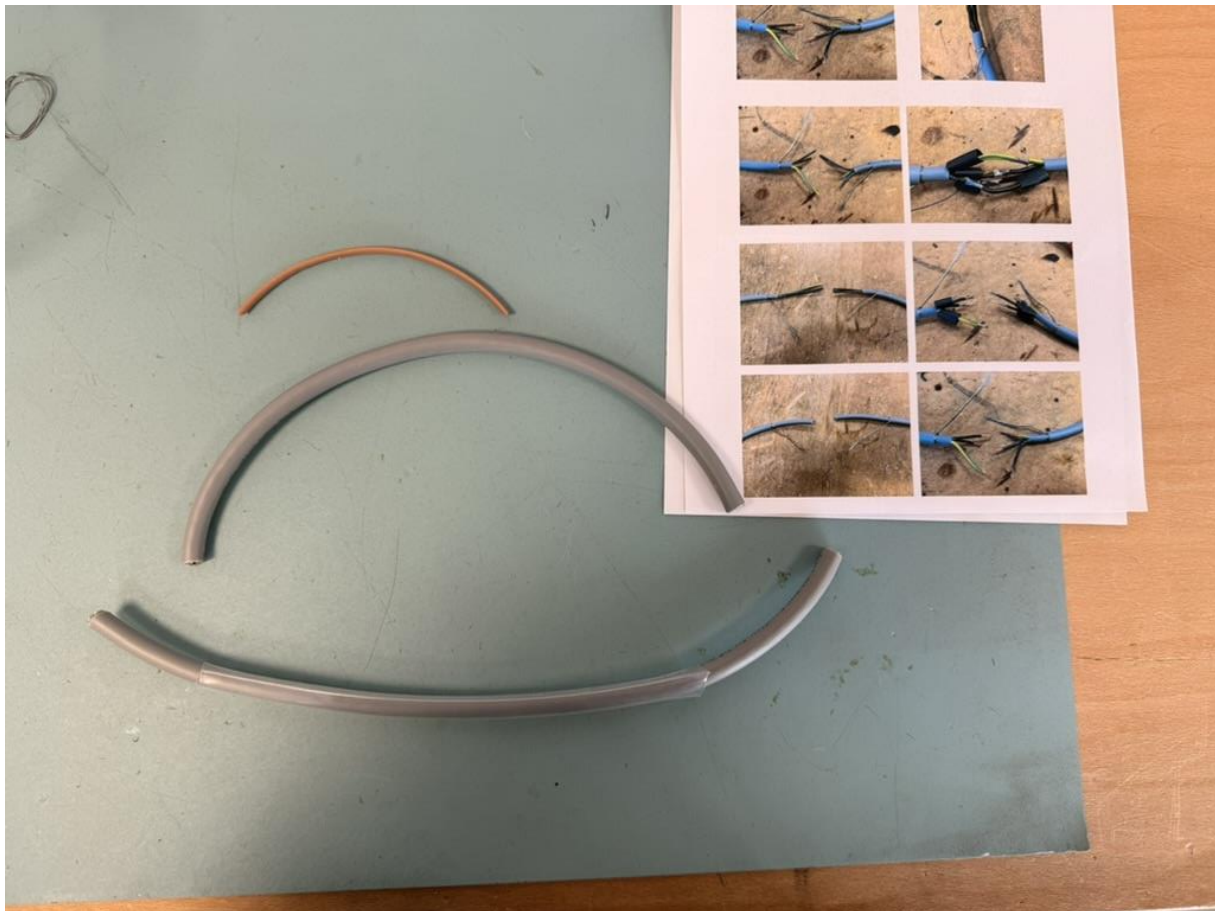


Figure 2: Cable cut in half with transparent shrinking tube

Stripping both ends off

Here, I used a wire stripper tool to carefully remove the thick grey isolation housing of the cables itself.

What I needed to pay attention too is the pressure I used to strip the cable, if I would have done this wrong, in the best case, I would not be able to strip the cable, or in the worst case, I would damage the ground braid and potentially the wires too.



Figure 3: Wire Stripper that stripped both cable ends

Twisting the ground braids

This step was nothing to worry about, I just had to twist both ground braids in order to avoid having a messy work, and so that I would later on solder them easily.

I just had to pay attention by the facts that I could not put too much force into the twisting to avoid them to break.

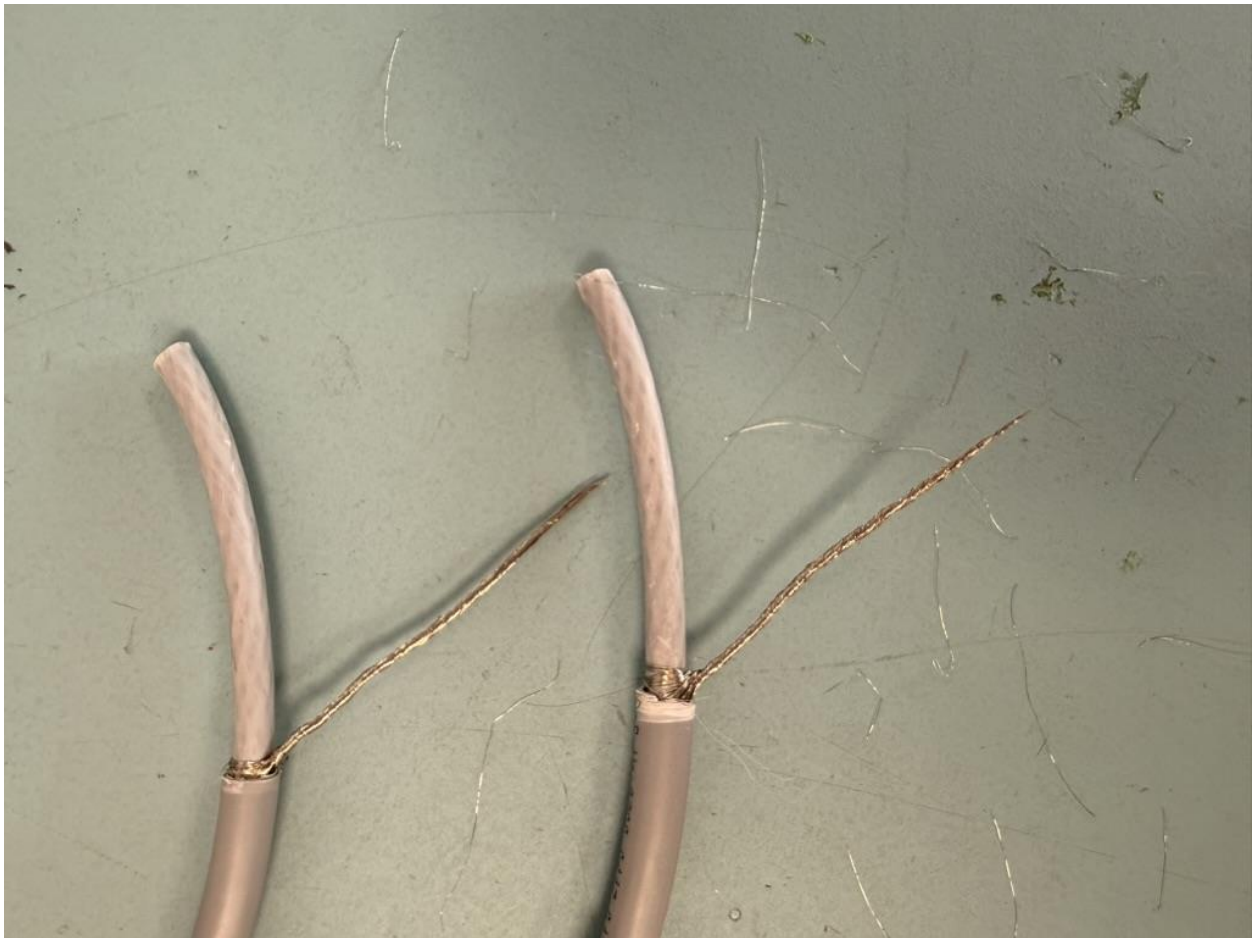


Figure 4: Twisted Ground Braids

Stripping the second layer of the grey isolation

For this step we used exactly the same technique as the primary grey isolation layer.

So again, just had to be careful on the pressure used to strip both isolations.



Figure 5: Both Grey Layers stripped exposing the isolated wires of both cables

Stripping the isolated wires

This is the preliminary step of the soldering.

Here, I had to carefully strip all wires to expose their copper.

The copper wires are fragile, if I put too much pressure using the Pressing Plier tool, I would also cut the whole wire off instead of the isolation.

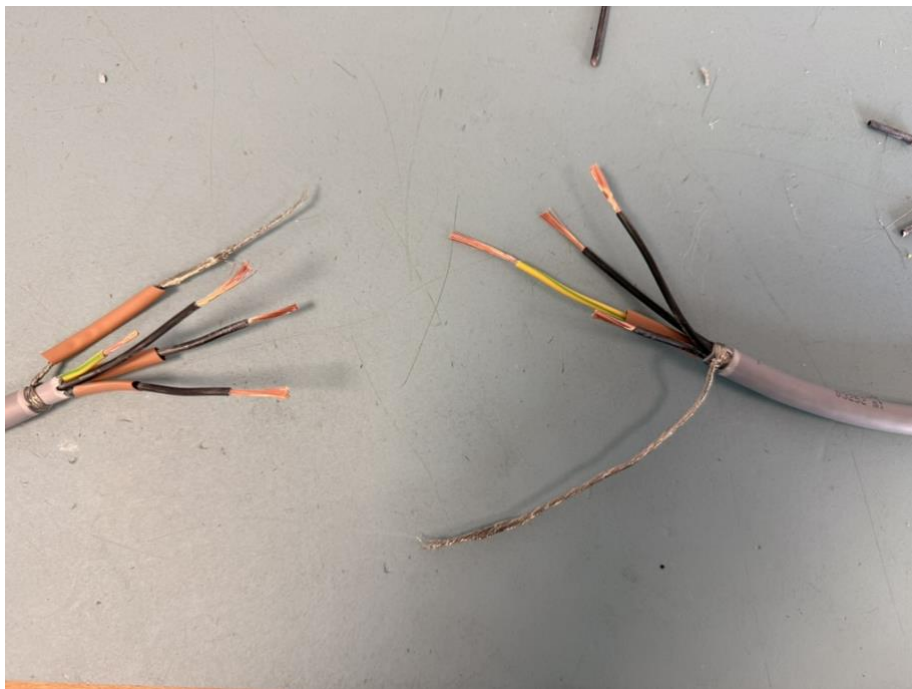


Figure 6: Cables with exposed copper wires

Putting soldering iron on the exposed wires

This step was for me the easiest, first you put the brown shrinking tubes on every wires, on one end only, then you would just put soldering iron on the copper.

What I had to pay attention, is that the soldering iron is obviously very hot for the shrinking tube, so I had to make sure that the shrinking tubes were far away to proceed without any of the tubes to shrink.

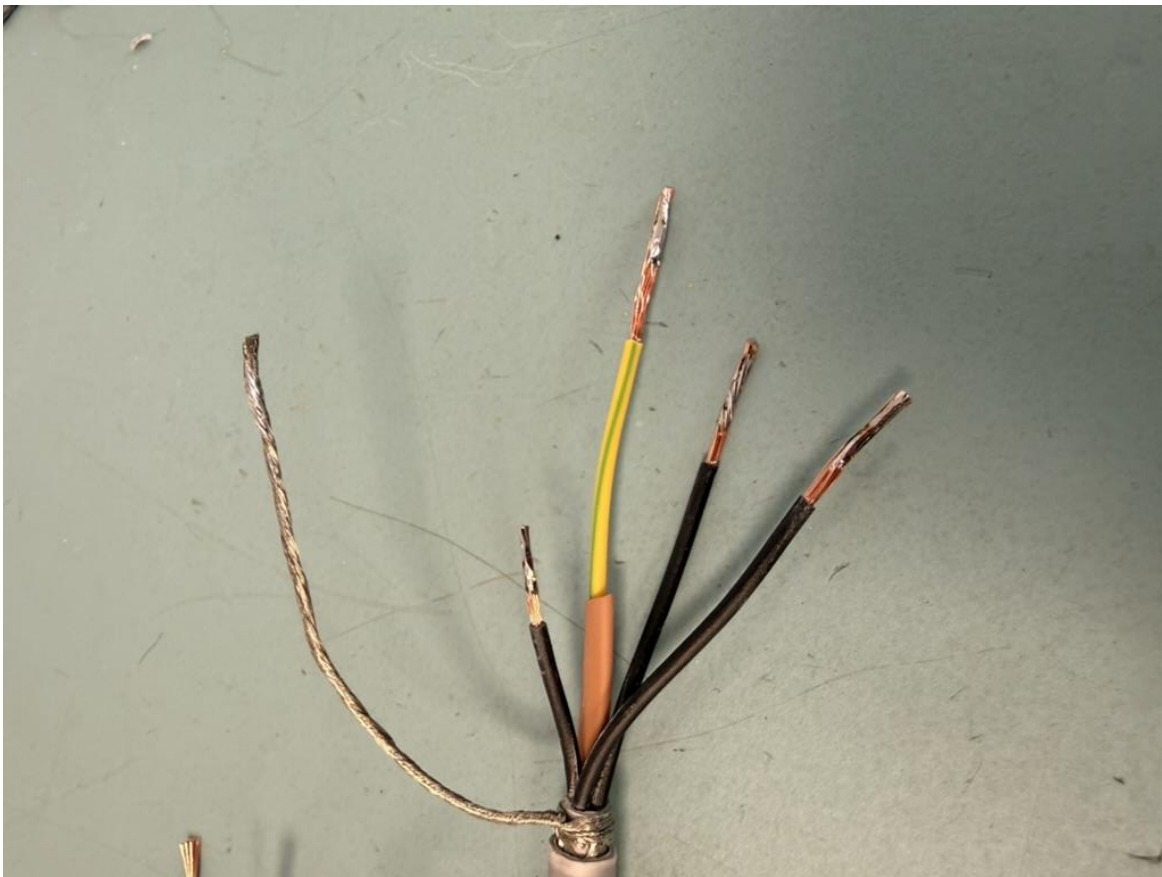


Figure 7: Exposed copper wires with soldering iron soldered to

Soldering both ends accordingly

The key to have a nice result, is that I have cut every ends on a different length, meaning that we would not result by having a big chunk of shrinking tubes on one place, to solder the wires to the other end, simply melt the already melted soldering iron we have put in the last step so that both wire ends would fusion together, and add a small amount of soldering iron too to guarantee the hold. As you can see on this image (Figure 8), every shrinking tubes are barely touching themselves since the wires are all cut to different lengths.

This was mainly my main concern, to avoid having a big chunk of shrinking tubes on one place.



Figure 8: Successfully soldered cable

What I could improve

I said in this User Story that I paid attention to carefully cut it as much as possible in half, but somehow my brain did not manage to do it at all, I had more or less like 60%/40% cut instead of a 50%/50%, I guess I did not pay attention enough in this step.

I know that I could get a better result if I would space more the cutting points of the wires, I will next time improve it.